

# Software Studio

## 軟體設計與實驗

# Assignment 02

## Web Mario

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# Goal

- Complete a “**Mario style game**” by **Cocos Creator**.
- You can use the materials TAs provide or download needed materials from some open-source webpages to beautify the appearance.
- Report which items you have done (items in scoring page) and describing other functions or feature in REABME.md.



# Mario

- Example result



# Scoring

- **Complete Game Process (5%)**
  - Start menu
  - Level select
  - Game view(including **game start / game over**)
  - You need to control the game process according to current game & player status



# Scoring

- **Basic Rules (50%)**

- World Map: (10%)

- The world must have **correct physics properties**,  
Ex: objects fall due to gravity, two different objects should collide with each other correctly, etc.
    - Background & Camera should **move according to the player's position**.
    - At least **1** world maps.

- Level Design: (5%)

- The scene should have “**Static**” wall
    - The scene should include question blocks which can **interact with the player**



# Scoring

- **Basic Rules (50%)**

- Player: (15%)

- Player should have correct physics properties.
    - User can control the player to **move and jump** by keyboard.
    - When player touches enemies or be attacked by enemies, it will **get hurt** or the number of its life will decrease.
    - When player get **out of the bounds**, the number of its life will decrease.
    - When player dies, it can **reborn** at the initial position.



# Scoring

- **Basic Rules (50%)**

- Enemies: (15%)

- Enemies should have correct physics properties.
    - **At least 1** type of enemies.
    - Only when player hits on their heads can kill them.

- Question Blocks: (5%)

- At least **1** type of blocks (**super mushroom** which makes Mario bigger).
    - Blocks can interact with player correctly.



# Scoring

- **Animations (10%)**
  - Player has walk & jump animations (5%)
  - Enemies Animation (each for 2%, up to 5%)
- **Sound Effects (10%)**
  - At least one BGM (2%)
  - Player Jump & die sound effects (3%)
  - Additional sound effects (each for 1%, up to 5%)
  - **All sound effects can't stop BGM**
- **UI (10%)**
  - Player life (3%)
  - Player score (5%)
  - Timer (2%)
- **Appearance(subjective) (10%)**





# Scoring

- **Bonus (At most 10%)**
  - Firebase (5%)
    - Deploy to Firebase page
    - Membership mechanism
      - Sign up, Login with firebase, save/restore game progress
  - Leaderboard (5%)
  - Multi-player game
    - Set up another backend server for online version (10%)
    - Offline version (5%)
  - Others...?



# Scoring

- **Git (5%)**
  - Use Git to do version control
  - Commit regularly (not just on the last day!)



# Scoring

| Item                          | Score |
|-------------------------------|-------|
| Complete Game Process         | 5%    |
| Basic-Rules                   | 50%   |
| Animations                    | 10%   |
| Sound Effects                 | 10%   |
| UI                            | 10%   |
| Appearance (subjective)       | 10%   |
| Bonus (describe on README.md) | 10%   |
| Git                           | 5%    |

**Total score is at most 110%**



# AI Usage

If you utilize AI tools during development, you must include a report titled **AI\_reference.pdf** in the **root directory** of your project. The report must include:

1. **AI Tool(s) Used:** Specify the model (e.g., ChatGPT, Gemini...)
2. **Scope of Usage:** For every segment of code generated or assisted by AI, provide:
  - Location: File name and specific line numbers (e.g., App.js, lines 45-82).
  - Prompt & Response: The exact prompt you used and the AI's output (Screenshots are highly recommended).
  - Refinement & Explanation: Show your modified version of the code and provide a brief explanation of why you made those changes and how the logic works.
3. **Statement of Non-Usage:** If you did not use any AI tools, simply state: "No AI tools were used in this assignment." inside the PDF.

We strongly suggest that you thoroughly analyze any code produced by AI and rewrite it using your own logic, in case that it may be flagged as plagiarism regardless of the documentation.



# Reminder

- Deploy your web page to Firebase page, and ensure it works correctly.
- Upload all source code to FTP.
  - .html or .css, .js, **.ts**, etc.
  - source files
  - **README.md**
  - Zip all files into **Assignment02\_學號.zip**  
(Ex: Assignment02\_109062000.zip)
- **FIRM deadline: 2026/05/28 23:59 (commit time)**
- **MD5 checksum (if you didn't do this → -10%)**
- You will be **graded ZERO** under one of the following conditions
  - Missing the deadline, no matter what kind of reasons you have...
  - Plagiarism (抄襲), both the plagiarist (抄襲者) and accomplice (被抄襲者) will be graded zero.
  - Crashed program (e.g., 404 - Not Found)
  - Not upload the source files to FTP



# Reminder

- Deploy your web page to **Firestore page**, and ensure it works correctly. (if you didn't do this → 0%)
  - Your main page should be named as “index.html”
- Upload source code to FTP.
- Compress files into **Assignment02\_學號.zip** then upload
  - index.html, .css, .js, README.md, etc.
  - **Do not add node\_modules into zip file.**  
**(-5 if you upload node\_modules)**
  - If you upload the files again, please change the filename become **Assignment02\_學號\_v?.zip** (v? -> which version)
- **MD5 checksum** (if you didn't do this → -10%)
- **FIRM deadline: 2026/05/28 23:59** (commit time)
- Upload your **MD5** and **web link** and **GitHub / GitLab Url** to eeclase



# 如何不被判定抄襲？

- 參考相關網站的程式碼、透過生成式AI產生的程式碼，會因為來源相同、prompt相近導致被Moss判定為抄襲（例如function名稱相同）。
- 因此建議先理解網站\生成式AI\好友討論所產出的程式碼，再用自己的邏輯重寫，以降低被判定為抄襲的可能性。
- 若已知自己有涉及上述的手段來寫作業，建議先用MOSS比對自己的程式碼與參考的資料來源的相似度，確保比例低於20%，以降低被判定為抄襲的可能性。
- 關於Moss的申請與使用，可以參考下面的連結。  
<https://hackmd.io/@GjP3NF8qT0SXa2TjRxSKeQ/HJvqlGerE>





# 作業繳交規則

- 請**務必**透過 MD5 獲得作業checksum 後填入 google 表單。
- 如遇各種原因無法在作業期限前完成上傳，我們將比對 checksum。若 checksum 一致則不算遲交。
- **繳交作業以MD5為主**，若有多個 checksum 則**取時間最晚**的為主。
- [MD5 online generator](#)
- [MD5 checksum 登記表單](#)
- [MD5 checksum 登記查看](#)
- [MD5 使用方法](#)





# 作業繳交流程

請務必遵守以下規則：

1. 將作業打包壓成zip檔，檔名格式以及打包內容以當次作業公告為主。
2. 幫zip檔產生MD5，並填寫google表單。
3. 將zip檔上傳ftp。
4. 將MD5繳交至eeeclass。

**若違反任何一項繳交作業SOP，  
一律扣作業總分10分！！！！**

**貼心提醒：**請同學守護zip檔至作業成績公告或期末。

